

AI-Driven Survival Prediction in Metastatic Prostate Cancer:

Integrating PSA Kinetics and Genomic Data with Classical and Deep Learning Models

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Background

Metastatic prostate cancer (metPC) remains a leading cause of cancer-related mortality among men in the United States. While metPC is clinically heterogeneous, clinicians need early, and individualized risk estimates to guide surveillance and treatment changes. Previous results suggest that genomic features and early prostate-specific antigen (PSA) kinetics carry strong prediction ability for clinical outcomes⁴. While the traditional Cox Proportional Hazards (CPH) model is the cornerstone of survival analysis, it has inherent limitations in handling time-dependent covariates such as serially measured PSA levels, a dynamic indicator of disease progression. Our goal is to assess whether AI-based models can better harness the predictive power of these modality datasets.

Methods

6,457 men from 7 pivotal, randomized, multinational trials (TITAN, LATITUDE, COU-AA-301, COU-AA-302, ACIS, MAGNITUDE, GALAHAD). PSA measured at baseline and 4 monthly follow-ups; gene expression profiled for 605 patients (A302, ACIS, Titan)^{1,2,3}.

Features Assessed

- 12 Baseline clinical and demographic variables selected from AdaLASSO and the Halabi Model (2014)⁵ with appropriate transformations (i.e. race, ECOG baseline, hemoglobin baseline, platelets, serum albumin, bone/soft tissue/lymph node/visceral liver and lung metastasis, ALT, LDH baseline, ALP baseline)
- Aggregated genomic risk score from microarray gene expression data (from top 50 genes selected by random forest importance)
- PSA kinetics, as both a composite PSA score and longitudinal time-series at baseline + four follow-up timepoints

Modeling Approaches

- Classical CPH: with baseline and kinetic PSA, ± genomics
- DeepSurv⁶: Previous STOTA Deep learning adaptation for survival models
- Dynamic-DeepHit⁷: recurrent/MLP net for static + sequential features for survival models

Evaluation Metrics

- Concordance Index (C-index): Discrimination
- Integrated Brier Score (IBS): Calibration
- Time-dependent AUC (tAUC) for annual follow-up times

Results

Table 1. Study Population

Characteristic	N = 575 [†]	Characteristic	N = 575 [†]
Race		Bone Metastasis	
WHITE	527 (92%)	0	72 (13%)
ASIAN	29 (5.0%)	1	503 (87%)
BLACK	11 (1.9%)	Soft Tissue Node Metastasis	
OTHER	8 (1.4%)	0	286 (50%)
ECOG Performance		1	289 (50%)
0	425 (74%)	Liver Metastasis	
1	150 (26%)	0	567 (99%)
Hemoglobin (g/dL)	0.79 (0.73, 0.84)	1	8 (1.4%)
Platelets (x10 ⁹ /L)	0.59 (0.49, 0.70)	Lung Metastasis	
Albumin (g/dL)	0.90 (0.86, 0.94)	0	527 (92%)
Alanine Aminotransferase (U/L)	0.49 (0.37, 0.70)	1	48 (8.3%)
Lactate Dehydrogenase (U/L)	0.77 (0.70, 0.88)	PSA Score	0.18 (-0.42, 0.65)
Alkaline Phosphatase (U/L)	0.70 (0.53, 1.01)	Gene Score	-0.80 (-1.42, -0.21)
[†] n (%); Median (Q1, Q3)		[†] n (%); Median (Q1, Q3)	

Figure 1. tAUCs across Feature Comparisons based on CPH models

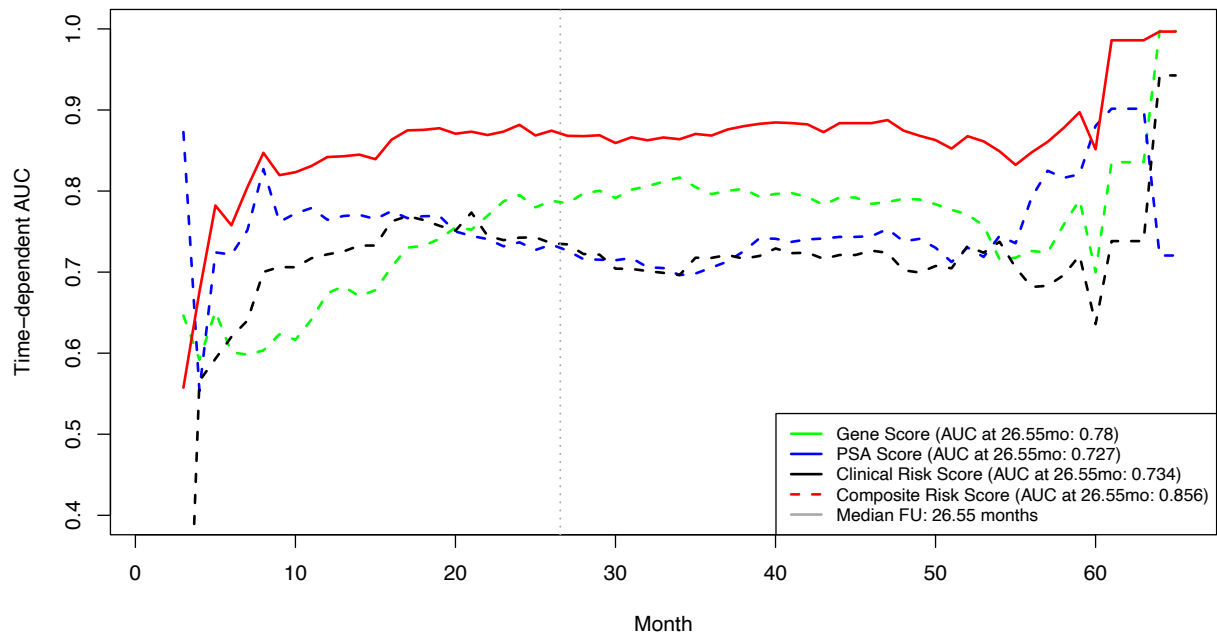


Figure 2. Forest Plot of Hazard Ratios from the best CPH Model

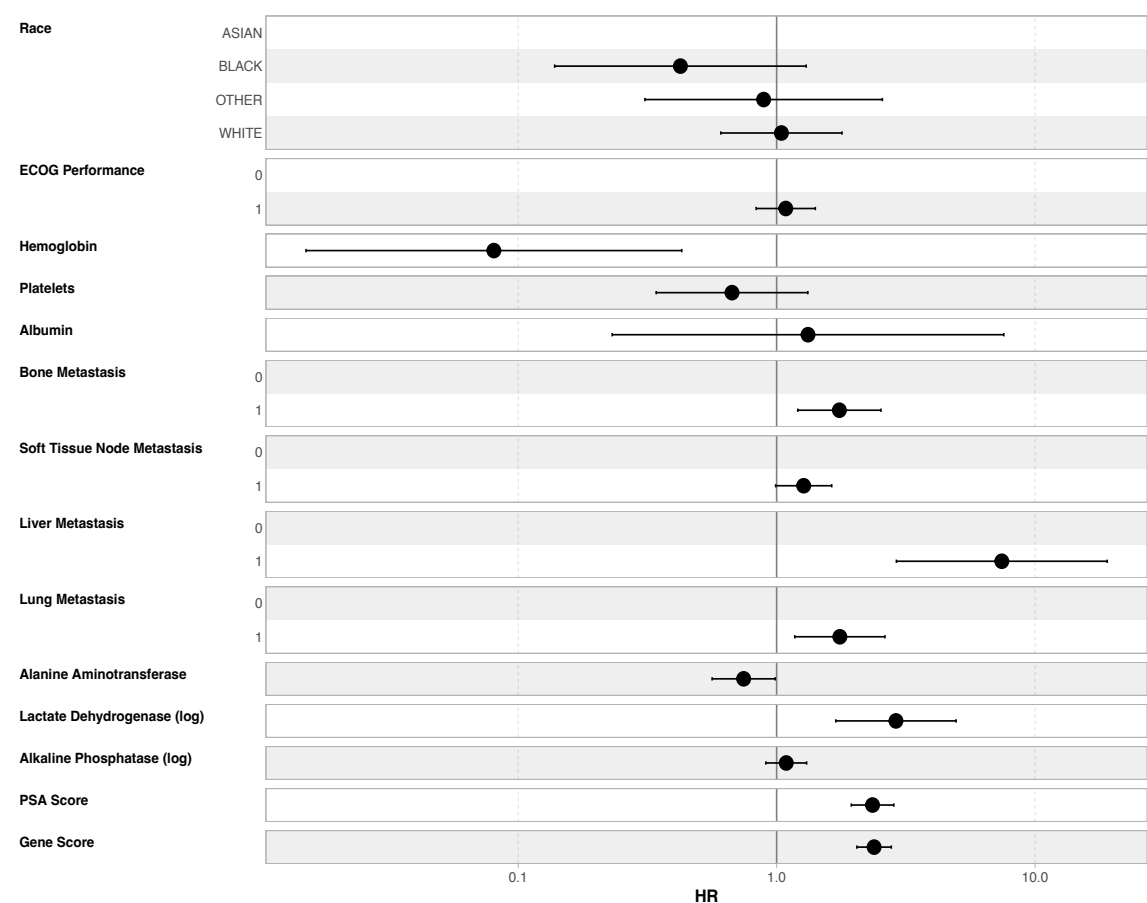


Fig 3. Model Performance Evaluation across Input Features and Statistical/Deep Learning Model

All permutations of input features – Clinical, Genomic, PSA - were combined to predict survival outcomes. The gray boxes indicate that the deep learning models were skipped due to too few input features.

Fig 3A. C-Index Heatmap. C-Index scores ranged from ~0.639-0.794, with higher values (increasing red) representing better discrimination prediction.

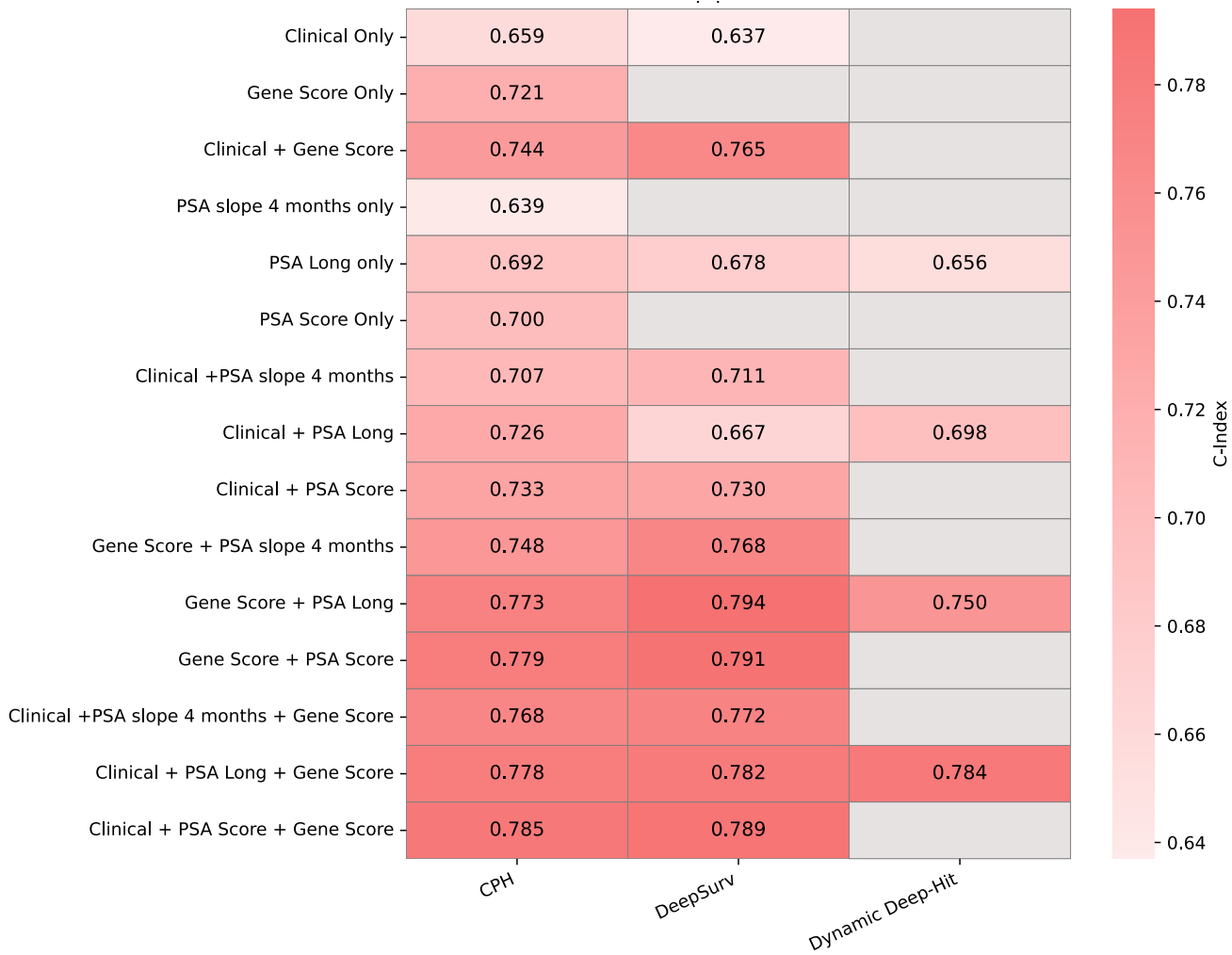
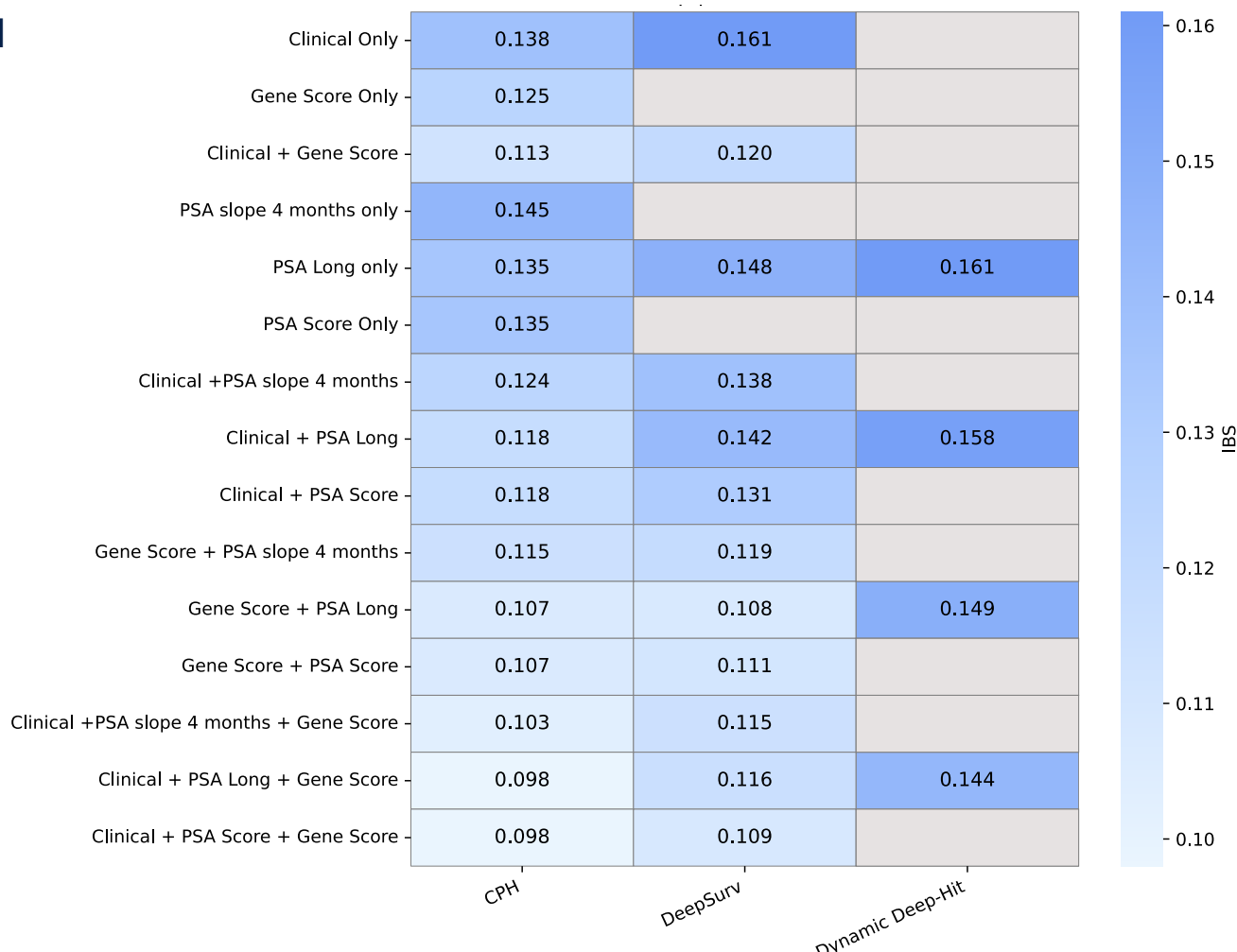


Fig 3B. IBS Heatmap. Lower IBS values (lighter blue) represent better calibration and model performance.



Conclusions

PSA and Genomics Additive Value

Both composite PSA risk scores and longitudinal PSA values improve model accuracy versus clinical variables alone (C-index = 0.654, IBS = 0.139).

Metrics Highlight:

CPH

- Best C-index: 0.785 (Clinical+PSA Score+Gene Score)
- IBS: 0.098
- Genomic score HR: 2.38 (95% CI: [2.04, 2.78])
- PSA score HR: 2.35 (95% CI: [1.94, 2.84])

DeepSurv

- Best C-index: 0.794 (Gene Score + PSA Long)
- IBS: 0.108

Dynamic Deep-Hit

- Best C-index: 0.784 (Clinical + Gene Score + PSA Long)
- IBS: 0.144

Key Findings

The results demonstrate that deep learning provides tools for integrating complex, time-dependent biomarkers in survival analysis. Specifically, longitudinal PSA measurements and aggregated PSA scores are more informative than PSA slope-derived static features. Incorporating PSA features alongside genomic scores substantially improved predictive accuracy across all models compared to clinical covariates alone, regardless of model approach.

Future Direction

This data exhibits sparse multimodality. Restricting analyses to patients with complete data across all modalities reduces statistical power and generalizability. Therefore, we aim to develop models that can handle heterogeneous data types with high rate of incompleteness. Our ultimate goal is to develop an AI- and LLM-enabled clinical decision support platform for patients with metPC.

Reference

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